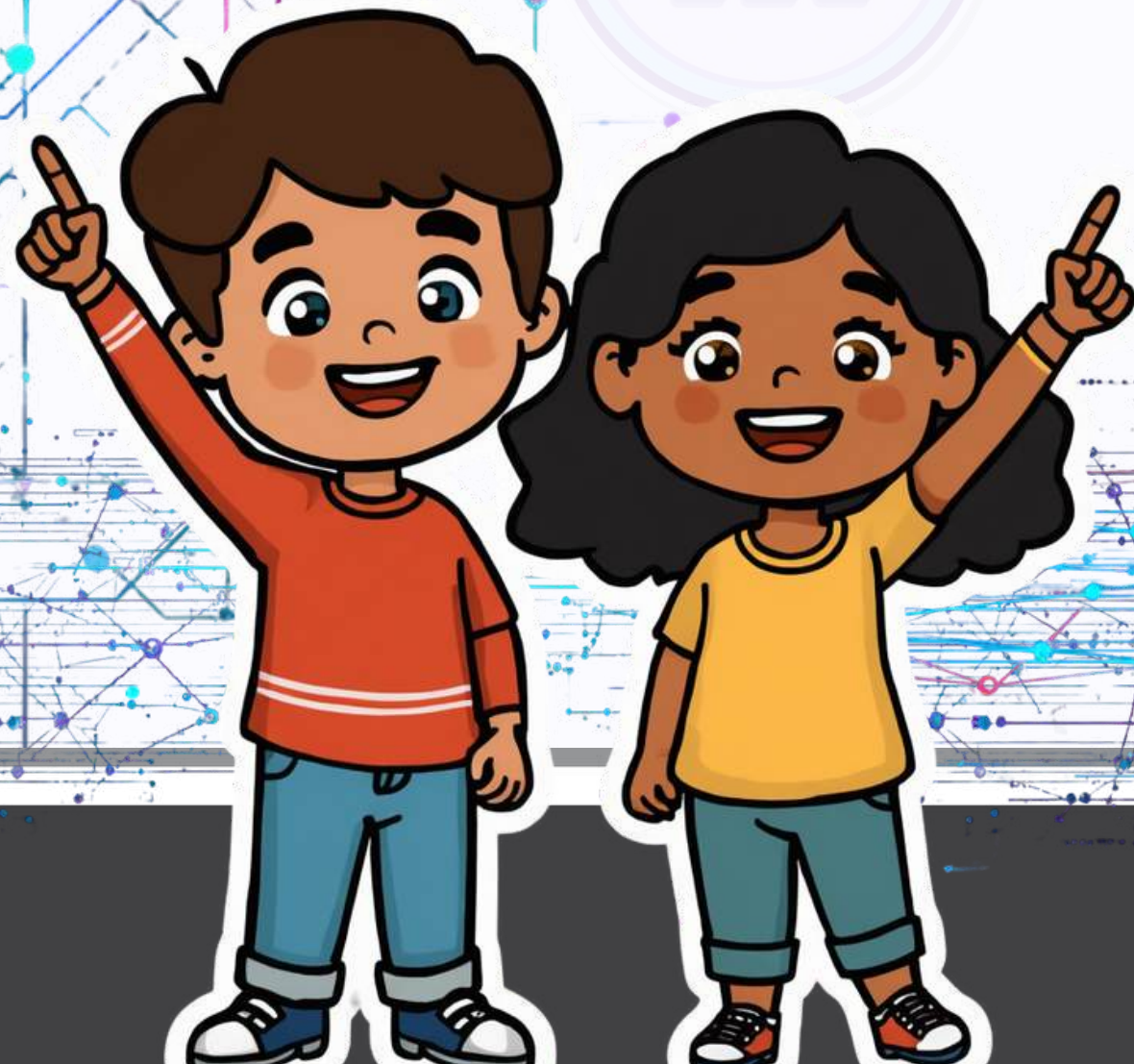
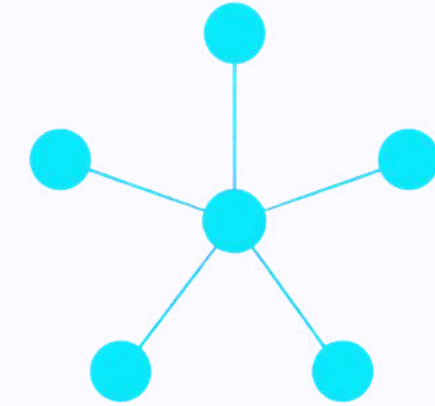




CONNECT THE DOTS:

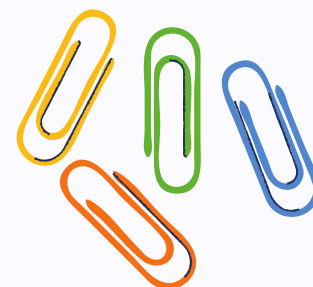
**NAVIGATING
NETWORKS
WITH BFS AND DFS**





How Are You Connected to a Celebrity?

How is it that you are connected to a celebrity through just a few friends? This is because we all live in a giant social network — a web of connections! Everyone you know is connected to someone else, and those people are connected to even more people!





What Is a Graph?



Family Tree

People connected to each other — parents, grandparents, cousins!

Metro Map

Stations connected by tracks — like Delhi Metro!

Friend Circle

You and your friends connected in a social network!

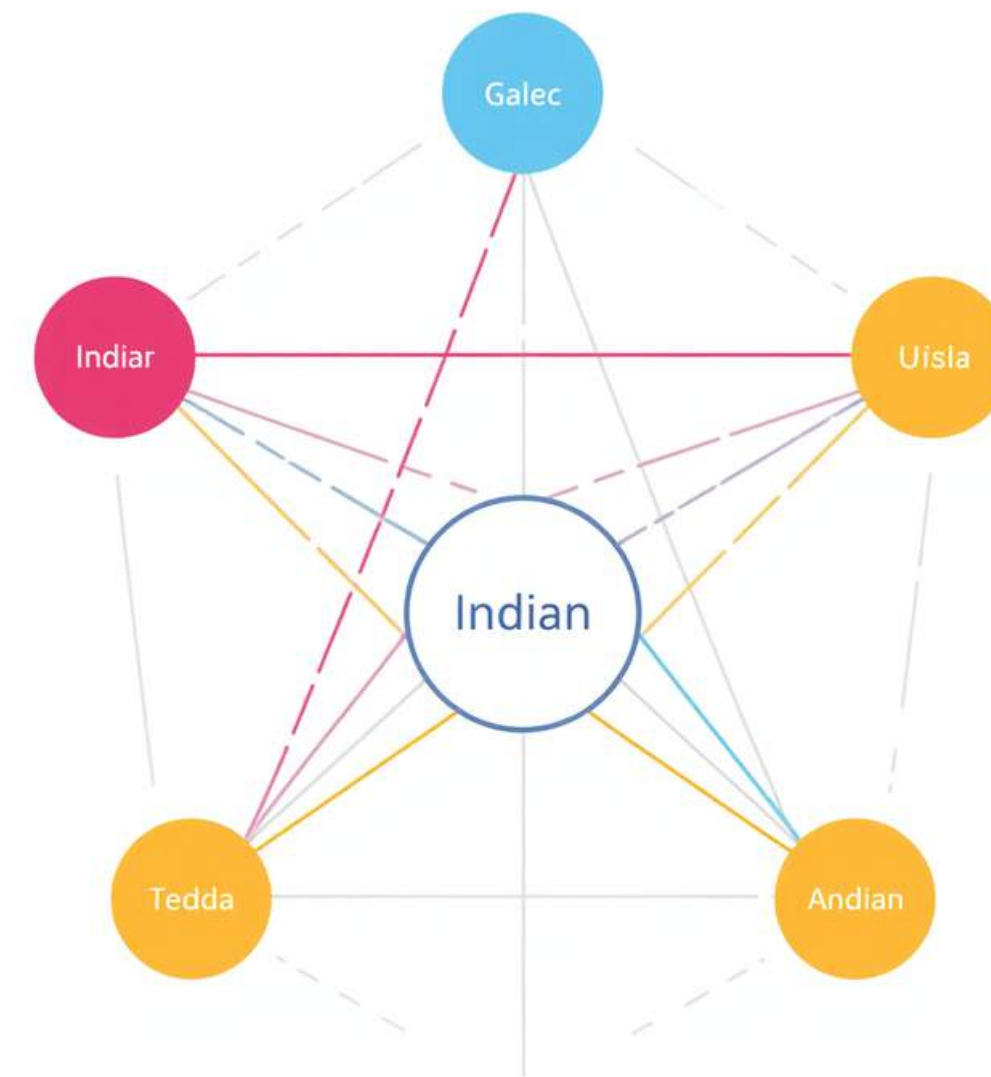




Nodes & Edges

Nodes are the dots or points in a graph — like metro stations, cities, or people in a network. Edges are the lines that connect these dots — like train tracks between stations or friendships between people.

- ✓ Nodes = Points (stations, people, cities)
- ✓ Edges = Lines connecting them (tracks, friendships)





Delhi Metro

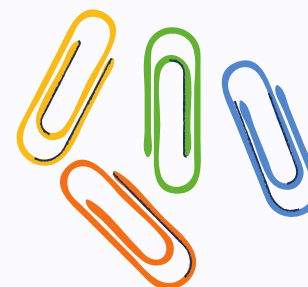
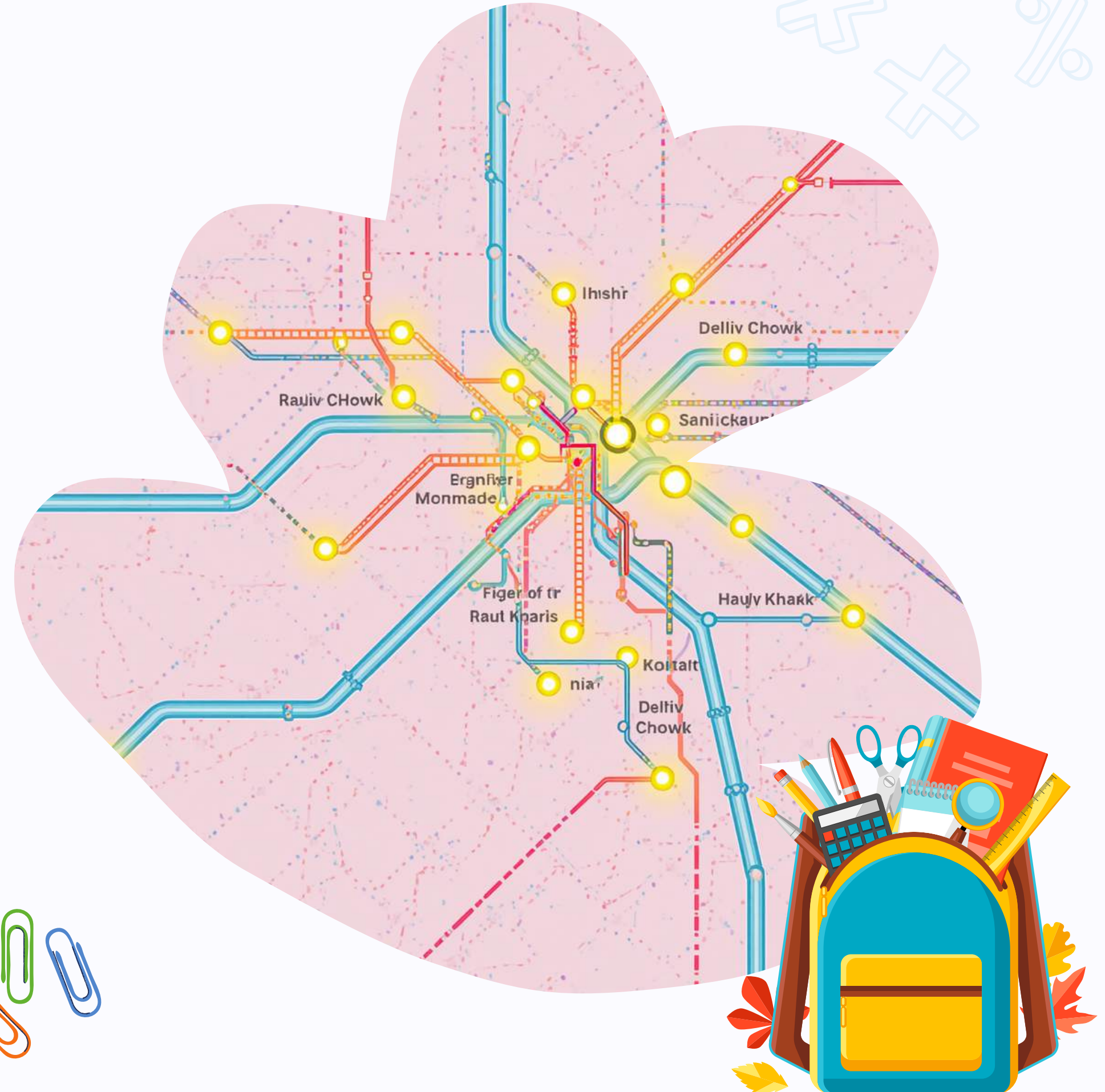
A Real Indian Graph

The Delhi Metro is a perfect example of a graph in real life!

Stations like Rajiv Chowk and Hauz Khas are **NODES** — the dots or points in our graph.

The metro tracks connecting these stations are **EDGES** — the lines that link nodes together.

Every time you travel from one station to another, you're moving along edges between nodes!





Meet Buddy BFS & Daring DFS!

Buddy BFS: "I like to check all nearby friends first before going farther." He's friendly, calm, and searches in layers like ripples in a pond.

Daring DFS: "I love to explore one path all the way to the end before trying another!" She's adventurous and goes deep into every corner.

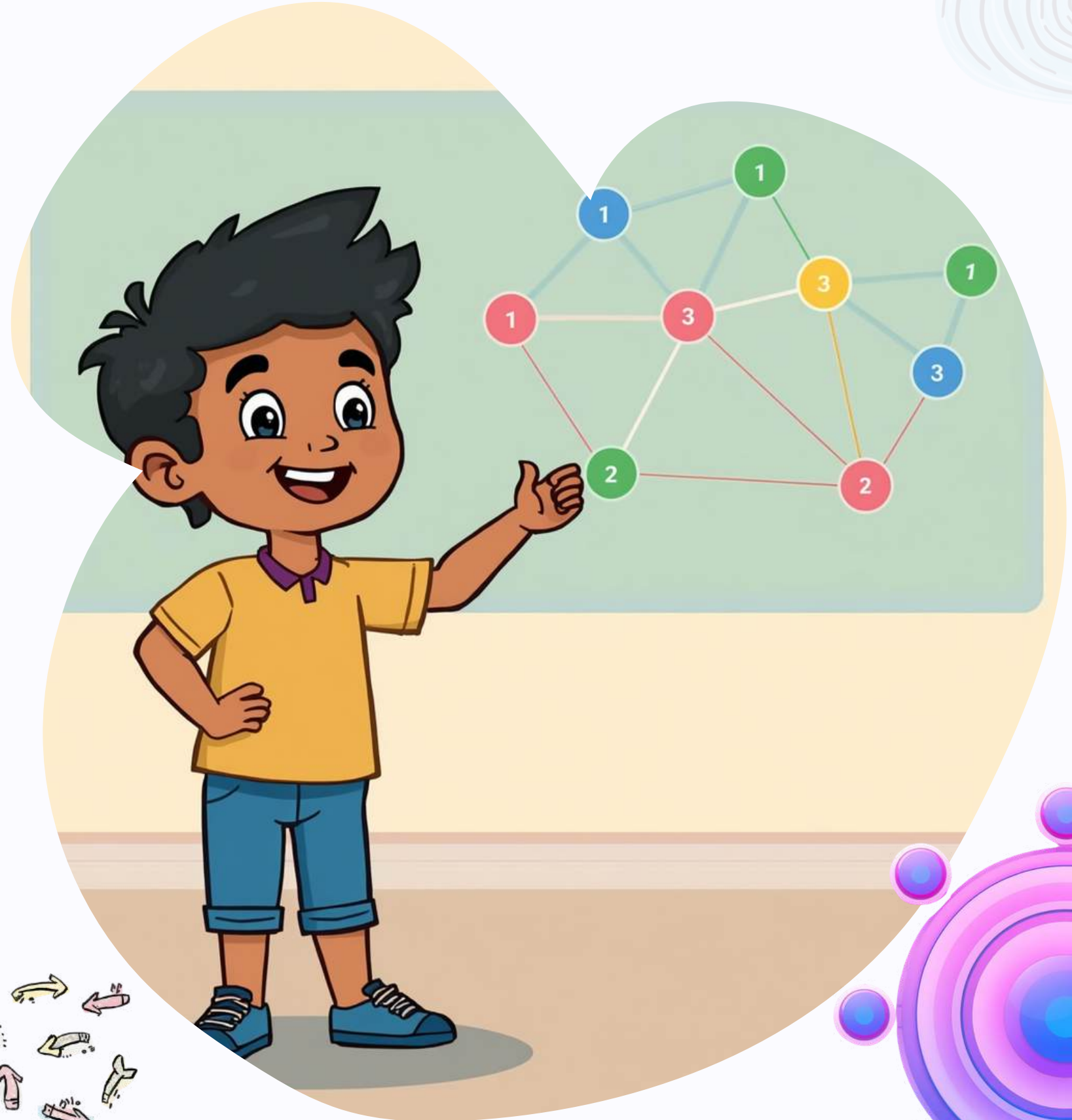




How Buddy BFS Searches

Like ripples in a pond — spreading out level by level. Buddy BFS visits all neighbors close by before moving outward.

First, check all nodes one step away. Then check all nodes two steps away. Keep spreading outward until you find what you need!

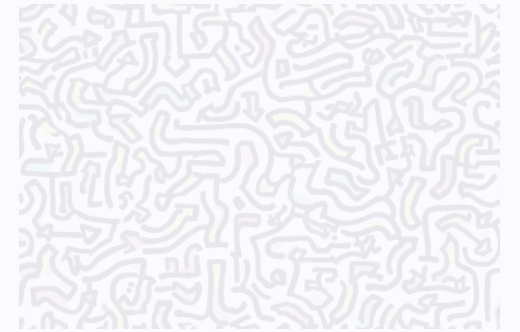




How Daring DFS Explores

Like exploring a maze by going as deep as possible down one path before trying another route.

Daring DFS follows one path all the way to the end, then backtracks to try another path. It's thorough and adventurous!





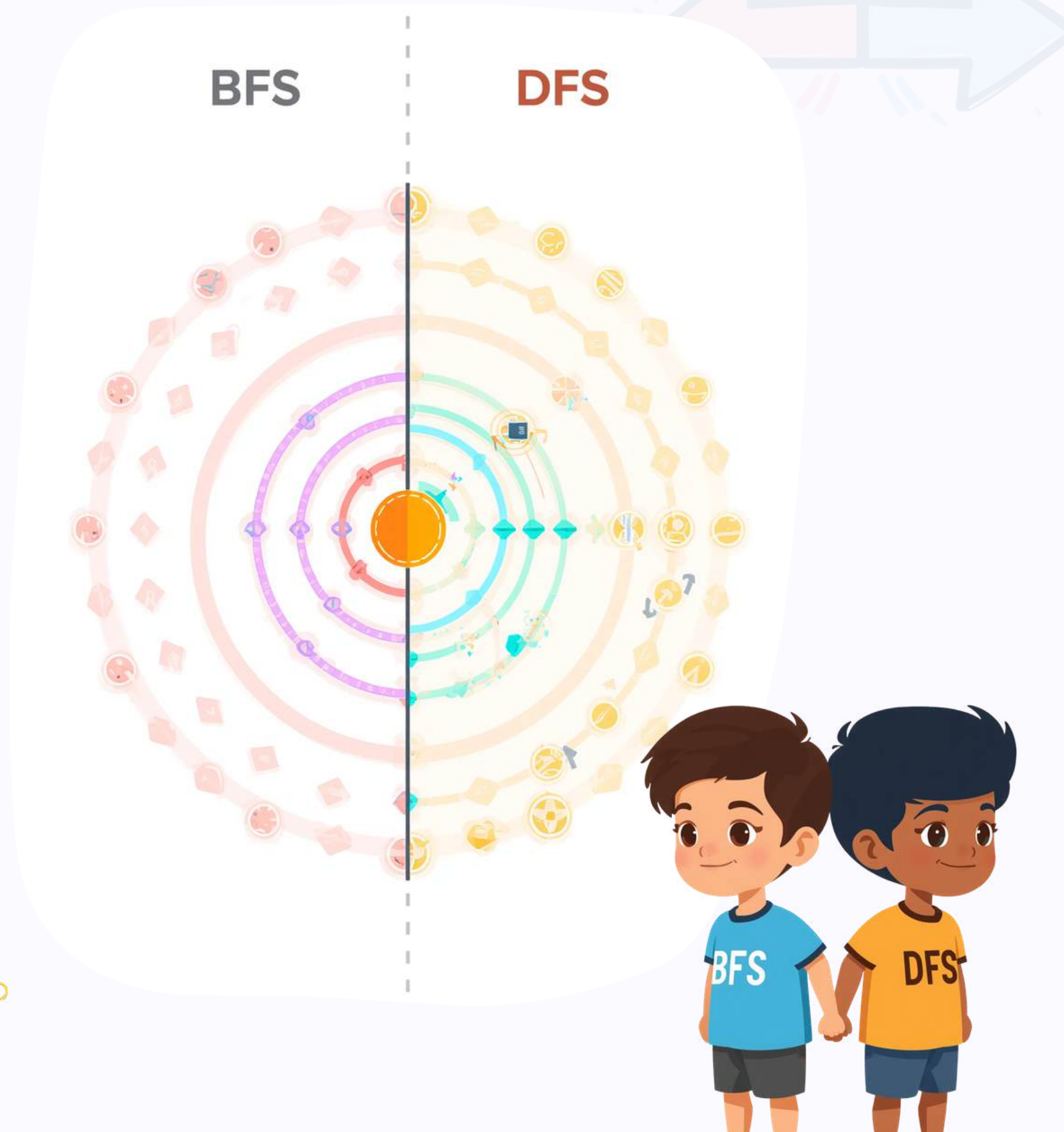
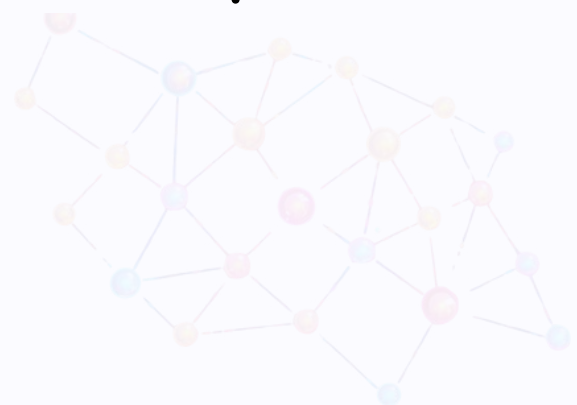
BFS vs DFS Compared

BFS (Buddy's Way):

- Nearby first - checks all neighbors before moving outward
- Searches in layers like ripples in a pond
- Finds the shortest path quickly
- Best when your goal is close by!

✔ DFS (Daring's Way): Deep first - explores one path all the way to the end

✔ Summary: Use BFS to find the closest place. Use DFS to explore every corner!





Activity Time: The Metro Rescue!

You lost your bag at a metro station! Which way will you search?

- BFS Way: Check all stations one stop away first, then two stops away, and so on.
 - DFS Way: Follow one metro line all the way to the end before trying another line.
- Which strategy finds your bag faster?

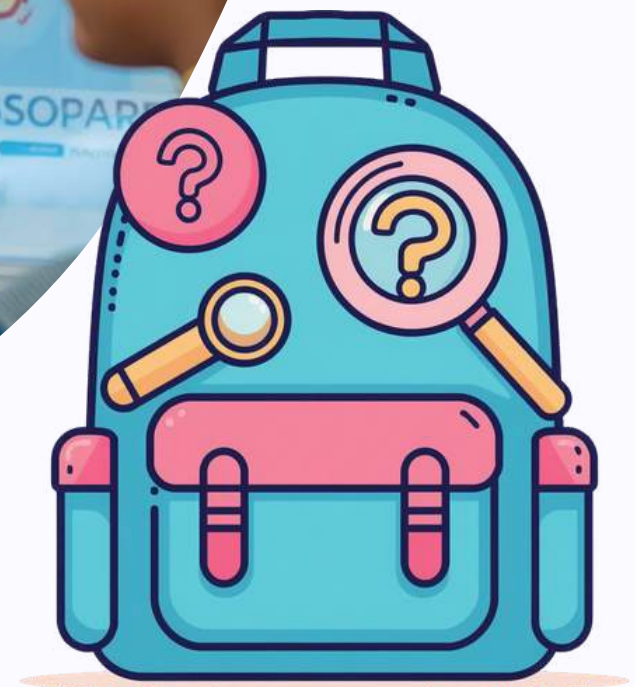
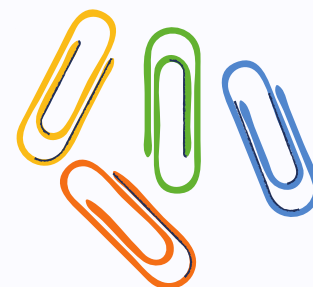




What Would Buddy BFS Do?

Buddy BFS checks all nearby stations first,
so he finds your bag quickly if it's close!

BFS finds the shortest path to your lost bag
by searching in layers — one stop away, then
two stops, then three!



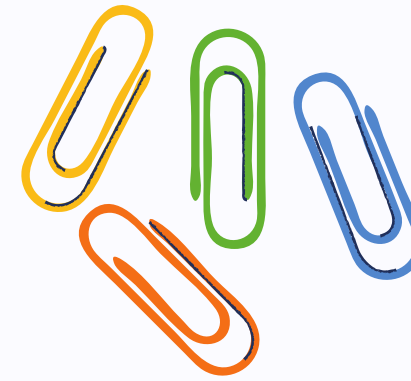


What Would Daring DFS Do?

Daring DFS follows one metro line all the way to the end. This might take longer if your bag is nearby but not on that path!

DFS explores deeply but may miss a nearby solution at first. It's great for thorough exploration, but not always the fastest way to find something close.





Wrap-Up: What Did We Learn?

Graphs

Graphs are made of dots (nodes) connected by lines (edges) — like metro stations and tracks!

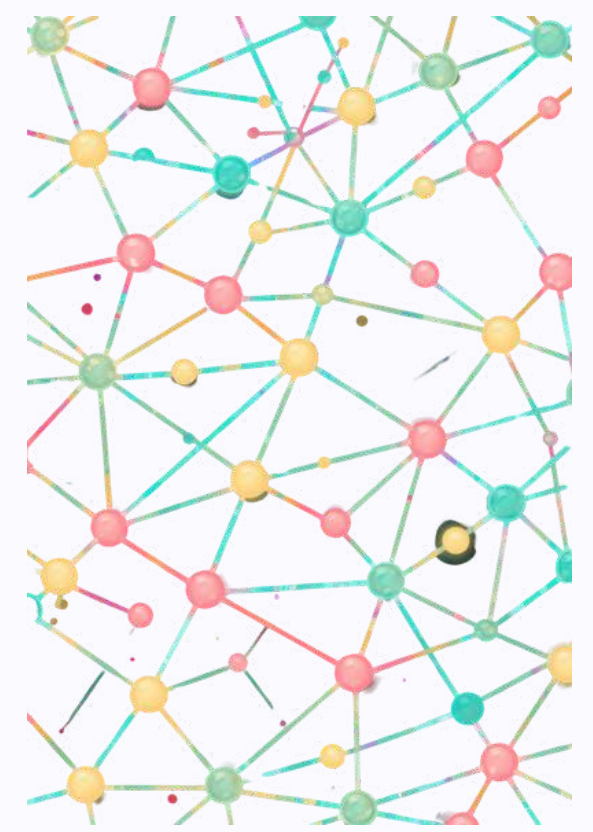
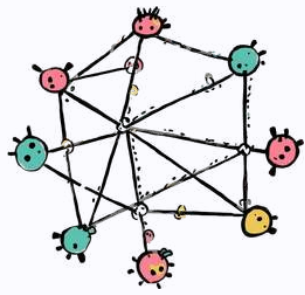
BFS Search

Buddy BFS searches in layers — checking all nearby places first to find the nearest things fast!

DFS Search

Daring DFS explores one path fully before trying another — great for thorough exploration!

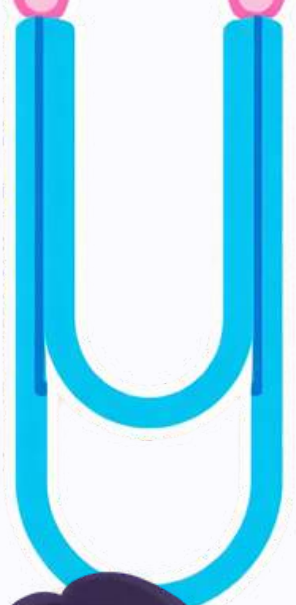




READY TO
NAVIGATE YOUR
OWN GRAPHS?

TRY IT NOW!





THANK YOU!

See You Next Time for More Graph Adventures!

Thank you for joining Session 1!

PLANRS Academy

